CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · DPP SHEET

Chapter 5: Quadratic Equations

Daily practice problems: 10 questions with an answer key

Try every question on your own first. The answer key is on the last page.

1. Check whether $x = -3$ is a root of $2x^2 + 5x - 3 = 0$.

2. If $x = \frac{2}{3}$ is a root of $3x^2 + kx - 2 = 0$, find k.

3. Solve $x^2 - 9x + 20 = 0$.

4. Solve $6x^2 + x - 12 = 0$.

5. Solve $\frac{x+1}{x-1} = \frac{3x-7}{2x-5}$.

6. Solve $x^2 - 5x - 10 = 0$, correct to two decimal places.

7. Solve $2x^2 - 4x - 3 = 0$, correct to two decimal places.

8. Find the nature of the roots of $4x^2 - 12x + 9 = 0$.

9. Find the value of k for which $kx^2 + 6x + 1 = 0$ has equal roots.

10. For which values of p does $x^2 + px + 4 = 0$ have real roots?

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Answer Key

1. $18 - 15 - 3 = 0$; yes
2. $\frac{4}{3} + \frac{2k}{3} - 2 = 0$, so $k = 1$
3. $x = 4$ or $x = 5$
4. $x = \frac{4}{3}$ or $x = -\frac{3}{2}$
5. It reduces to $x^2 - 7x + 12 = 0$, so $x = 3$ or $x = 4$
6. $x = 6.53$ or $x = -1.53$
7. $x = 2.58$ or $x = -0.58$
8. $D = 0$: real and equal, each $\frac{3}{2}$
9. $36 - 4k = 0$, so $k = 9$
10. $p^2 \geq 16$: $p \geq 4$ or $p \leq -4$

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